

Dashboard prototype that would assist lineup decisions based on the physical output from runs out of the batters box, located at [figures/readiness_dashboard.html](#).

I. Approach

a. Physical Output:

Many metrics exist to define the physical output during high speed runs. To narrow this down, a literature review showed that: 1. Total High Speed Running Distance ([Hader](#)) and 2. Acceleration events ([Varley](#)) were the most correlated with increased fatigue and strength decline in professional athletes day(s) after a match. Therefore, these metrics measure athlete load as a signal of next day readiness. The specific cutoffs on speed and acceleration were determined by player-specific metrics according to [Freeston](#) ("In-Game Workload Demands of Position Players in Major League Baseball").

Additionally, Statcast metrics such as Sprint Speed (peak 1-s window) and Burst distance are classically used to represent athlete performance which can indicate readiness along with load. With more history, ACWR (acute:chronic workload ratio) could layer on top but the current ~1-month window is too short for a stable chronic baseline.

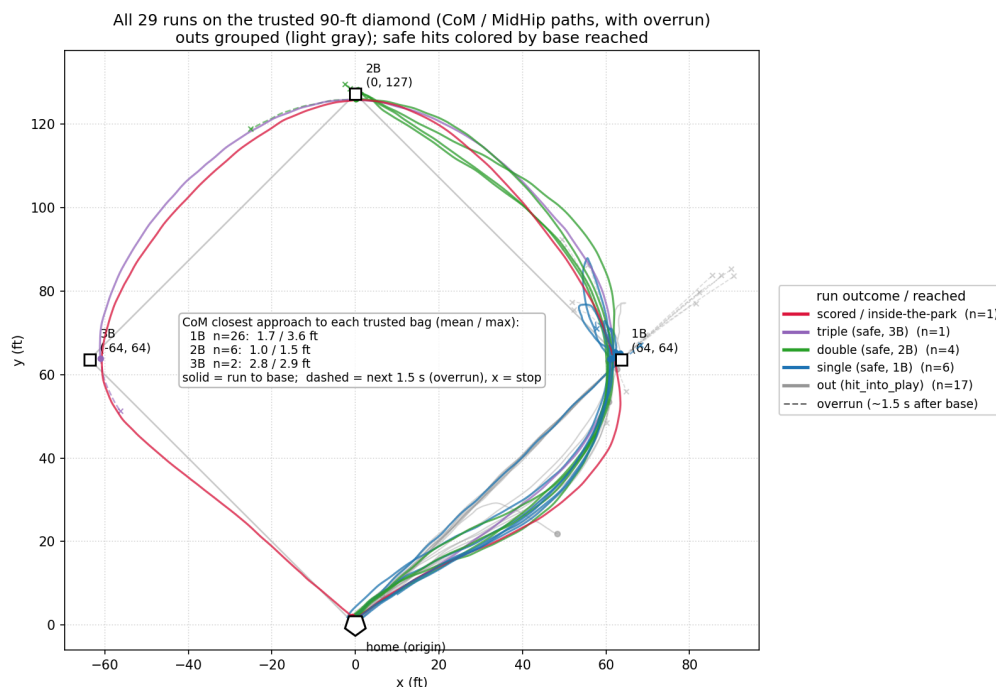
b. Player Readiness:

The idea is to watch whether the player's per-run output declines, game to game, relative to his own baseline based on a composite z score of these 4 metrics.

II. Data Processing

a. Categorizing Runs and Exploratory Visualization

See if a naive approach for locating bases can categorize Singles vs Doubles etc.:



- Assuming a perfect origin at home plate and MidHip as Center of Mass, negligible error
- Run outcomes were determined by the furthest plate the CoM came close to at the end
- Max sprints (going straight to first) vs failed double attempts (curved path to and back to first) are nuances to consider later on

b. Run-effort cluster analysis

An obvious confounder is that a slow run can also be game-driven (he's clearly out, or it's an easy double) not physical condition. So I clustered the runs by effort and filtered to the high-effort ones for analysis.

Clustered the 28 captured runs (dropped 1 of 29, the clip only caught 12 ft of travel) on five effort-shape features off the smoothed CoM (MidHip): Sprint Speed, Burst, path straightness, speed sustain, and braking decel (run_features.py). Left the run outcome out on purpose so it doesn't re-find singles vs doubles vs outs. Ward clustering (K-means + GMM as checks), silhouette → k=2 (cluster_runs.py).

Result: 24 high-effort vs 4 sub-max. The sub-max being 3 plays rounding the bases + 1 grounder.. ~86% of his runs are reasonably high-effort. Each run also gets a continuous effort score [0,1], so it's a soft weight, not a hard cutoff (run_clusters.csv).

Validation (cluster_runs.py): leave-one-out stability 97%, cross-method agreement (ARI) of 0.62, silhouette 0.49; peak speeds cap at 28.9 ft/s (Statcast range, avg 27). Also no slides detected (clips end at the bag).

c. Per-game output (game_readiness.csv, per_run_output.csv).

For each game: Sprint Speed, Burst, HSR distance, and time to 90% max velo were compiled, rolled into a composite output expressed as a z-score vs. his own baseline, effort-weighted, and carrying an n-qualifying-runs confidence flag plus recent-load context (days of rest, prior-7-day baserunning load).